



8.7.4 Wrap-Up: Cool Down

1. Rewrite the equation in vertex form by completing the square. Then, give the key features revealed by both standard form and vertex form.

$$y = -2x^2 + 12x + 1$$

Key Features

x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{1cm}}$				
right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{1cm}}$				



Unit 8, Lesson 8: Modeling With Quadratics in the Real World



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.

MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can interpret key features of quadratic relationships in a real-world context.
- ☐ I can interpret solutions as viable or non-viable options given a real-world context.
- ☐ I can represent constraints of a real-world context using set-builder notation.



8.8.1 Warm-Up: Error Analysis

1. Jarvis' work for rewriting the expression $4x^2 + 32x - 67$ in vertex form is shown. Camila told Jarvis he made a mistake. Find Jarvis' mistake and correct it.

$$(4x^2 + 32x) - 67$$

$$4(x^2 + 8x) - 67$$

$$4(x^2 + 8x + 16) - 67 - 16$$

$$4(x + 4)^2 - 83$$



8.8.2 Guided Instruction: Modeling With a Quadratic Relationship

Greenlings are a group of fish that share common features. The function

$G(x) = -10x^2 + 100x + 44$
models the number of greenlings caught, in thousands, in the United States, since 2006.



1. What does the variable x represent? Include the units.
2. What does the variable G represent? Include the units.
3. Find $G(2)$.
4. Interpret $G(2)$. Include units.

5. What is the y -intercept?

6. Interpret the y -intercept. Include units.

7. Rewrite the function in vertex form. Show your work.

8. Interpret the vertex in terms of the context. Include units.

9. Complete the table of values for $G(x)$.

x	-3	0	2	3	5	8	10	11
G								

10. Circle the values in the table that do not make sense in terms of the context.

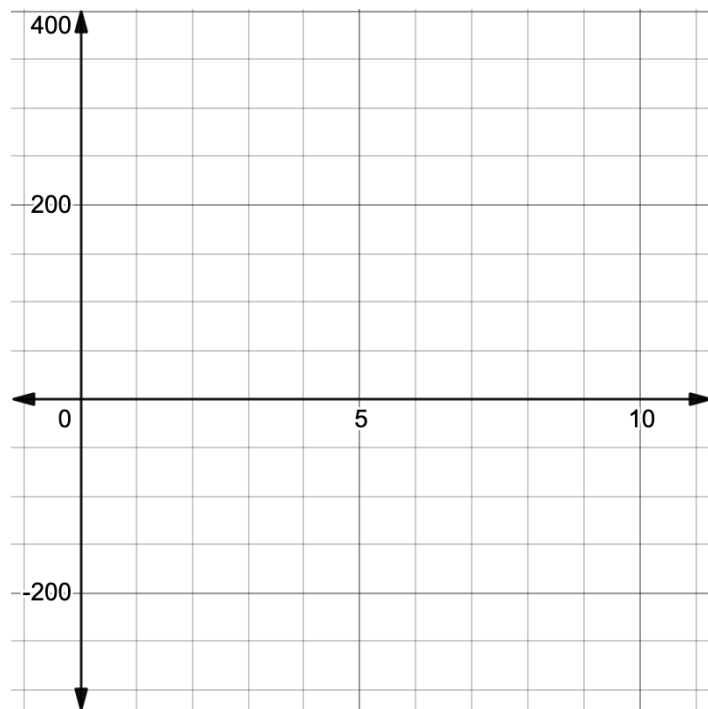
11. Explain why these values do not make sense.

12. Work with your partner to complete as many of the key features as possible using the given function $G(x)$, the function rewritten in vertex form, and the table of values. Do not use approximate or rounded values.

Key Features			
x -intercept		vertex	
x -intercept		increasing	
y -intercept		decreasing	
domain		range	
opens			
positive		left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$
negative		right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$



13. Graph the function.



14. Use the graph to estimate the x -intercepts.

15. Use the x -intercepts to determine when the number of greenlings caught reaches 0.

16. With your partner, discuss whether it's reasonable for the number of greenlings caught to be 0. Summarize your discussion.

17. Use the graph, table of values, or function to determine the reasonable constraints for the context. Write the constraints in set-builder notation.

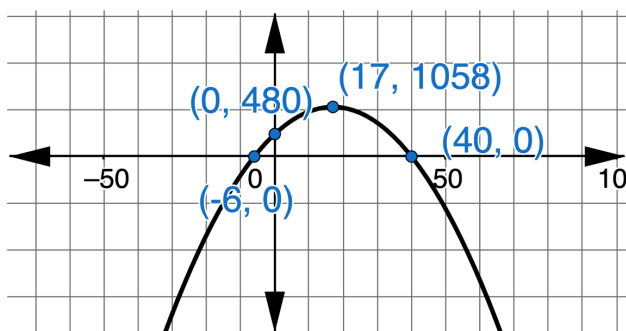
18. Discuss with your partner why these constraints may be different than the domain of the function $G(x) = -10x^2 + 100x + 44$.



8.8.3 Wrap-Up: Ticket Out the Door

1. A statistician gathers data over several years, starting in 1995, on the average number of trips a household makes to run errands per year. The statistician uses the data to find an equation that models the average annual number of trips for errands per household for the years since 1995. The statistician determined the function $E(x) = -2x^2 + 68x + 480$ models the situation.

The graph of the function is shown.



- a. Interpret $E(17)$. Include units.
- b. The x -intercepts are labeled on the graph. Determine how reasonable the x -intercepts are by explaining what they each mean in terms of when the average annual number of trips for errands per household will reach 0.

Unit 8, Lesson 9: Using Key Features to Solve Problems



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.

MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve real-world problems modeling quadratic relationships by determining the necessary key feature, and the form of the equation that reveals it.
- ☐ I can interpret key features of quadratic relationships in a real-world context.